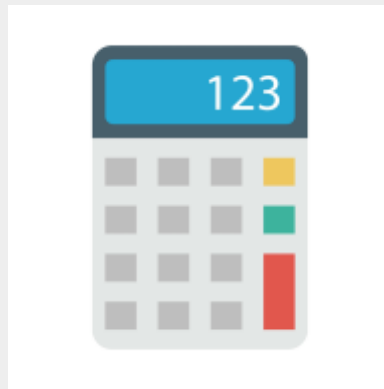


# Lecture 1

## Number System & Calculations



CSE260 : Digital Logic Design

## Objective

- Distinguish between analog and digital system
- Understand the advantage and limitation of digital system
- Understand the meaning of digital logic

## Analog vs. Digital

- Analog data can vary over a continuous range of values. *Example: speedometer*
- Digital quantities can take on only discrete values (0 and 1, high and low). *Example: Digital Computer, Decimal Digits, Alphabets*

# Digital System

- A digital system is a combination of devices designed to manipulate physical quantities or information that are represented in digital form.
- “A discreet information processing system”
- Signals: Discreet information

## Digital System Advantages

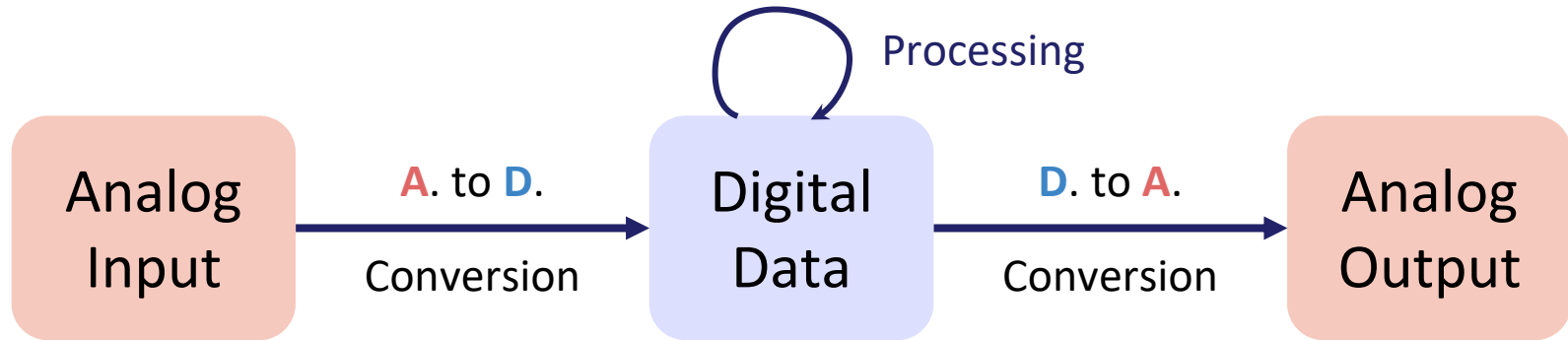
- Greater accuracy or precision
- Easier to design (generality)
- Easier information storage
- Programmability (instructions)
- Speed
- Economical

## Limitation

- The real world is mainly analog

# How to Overcome the Limitations

- Convert the real world analog input data into digital
- Process this digital data
- Then again convert into analog form



# Digital Logic

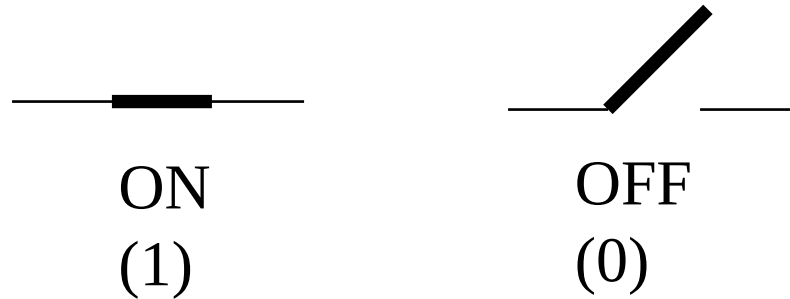
- Design logic is a term used to denote the design and analysis of digital system
- Digital logic is concerned with the interconnection among digital components and modules
- Digital logic design is engineering and engineering means problem solving

# Number systems and codes

- Digital Systems are built from circuits that process binary digits. BUT very few real-life problems are based on binary numbers
- So, a digital system designer must establish some **correspondence between the binary digits processed by digital circuits and real-life numbers**, events and conditions

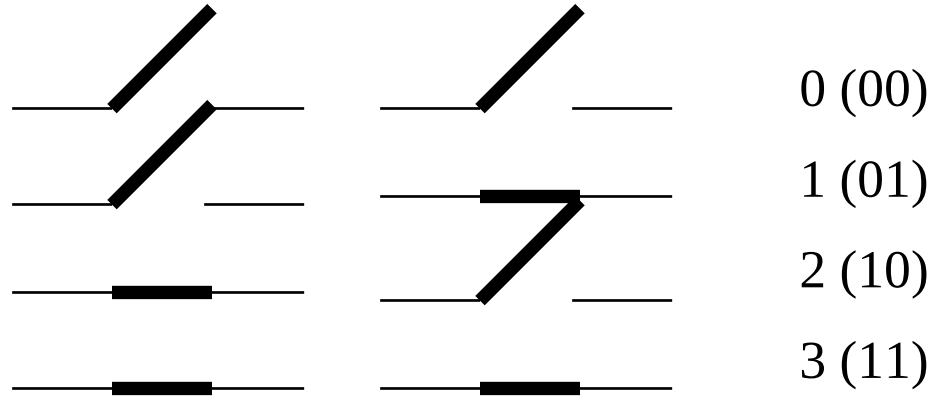
# Information representation

- Human decisions tends to be binary i.e. **Yes or No**
- Elementary storage units inside computer are *electronic switches*. Each switch holds one of two states: **on (1) or off (0)**.



# Information representation [cont.]

- We use a **bit** (binary digit), **0** or **1**, to represent the state
- Storage units can be grouped together to cater for larger range of numbers. Example: 2 switches to represent 4 values



# Information representation [cont.]

- In general,  $N$  bits can represent  $2^N$  different values.
  - 1 bit → represents up to 2 values (0 or 1)
  - 2 bits → rep. up to 4 values (00, 01, 10 or 11)
  - 3 bits → rep. up to 8 values (000, 001, 010. ..., 110, 111)
  - 4 bits → rep. up to 16 values (0000, 0001, 0010, ..., 1111)
- For  $M$  values,  $\lceil \log_2 M \rceil$  bits are needed.
  - 32 values → requires 5 bits
  - 64 values → requires 6 bits
  - 1024 values → requires 10 bits
  - 40 values → requires 6 bits
  - 100 values → requires 7 bits

# Positional Notations

- Decimal number system, symbols = { 0, 1, 2, 3, ..., 9 }
- Position is important

Example:

$$(7594)_{10} = (7 \times 10^3) + (5 \times 10^2) + (9 \times 10^1) + (4 \times 10^0)$$

- In general,

$$(a_n a_{n-1} \dots a_0)_{10} = (a_n \times 10^n) + (a_{n-1} \times 10^{n-1}) + \dots + (a_0 \times 10^0)$$

$$(2.75)_{10} = (2 \times 10^0) + (7 \times 10^{-1}) + (5 \times 10^{-2})$$

- In general,
- $(a_n a_{n-1} \dots a_0 . f_1 f_2 \dots f_m)_{10} = (a_n \times 10^n) + (a_{n-1} \times 10^{n-1}) + \dots + (a_0 \times 10^0) + (f_1 \times 10^{-1}) + (f_2 \times 10^{-2}) + \dots + (f_m \times 10^{-m})$

# Other Number Systems



- **Binary** (base 2): weights in powers-of-2. Binary digits (bits): ***0,1***
- **Octal** (base 8): weights in powers-of-8. Octal digits: ***0,1,2,3,4,5,6,7***
- **Hexadecimal** (base 16): weights in powers-of-16. Hexadecimal digits: ***0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F***

**Note: when base is  $r$ , coefficient values range from 0 to  $r-1$ .**

# Other Number Systems [cont]



- **Binary** (base 2): weights in powers-of-2. Binary digits (bits): **0,1**
- **Octal** (base 8): weights in powers-of-8. Octal digits: **0,1,2,3,4,5,6,7**
- **Hexadecimal** (base 16): weights in powers-of-16. Hexadecimal digits: **0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F**

| Binary | Octal | Decimal | Hexadecimal |
|--------|-------|---------|-------------|
| 0000   | 0     | 0       | 0           |
| 0001   | 1     | 1       | 1           |
| 0010   | 2     | 2       | 2           |
| 0011   | 3     | 3       | 3           |
| 0100   | 4     | 4       | 4           |
| 0101   | 5     | 5       | 5           |
| 0110   | 6     | 6       | 6           |
| 0111   | 7     | 7       | 7           |
| 1000   | 10    | 8       | 8           |
| 1001   | 11    | 9       | 9           |
| 1010   | 12    | 10      | A           |
| 1011   | 13    | 11      | B           |
| 1100   | 14    | 12      | C           |
| 1101   | 15    | 13      | D           |
| 1110   | 16    | 14      | E           |
| 1111   | 17    | 15      | F           |

# Decimal to Base-R Conversion

- ❖ Whole numbers: repeated **division**-by-R
- ❖ Fractions: repeated **multiplication**-by-R

# Repeated Division-by-2 Method

- To convert a **whole number** to **binary**, use **successive division by 2** until the quotient is 0. The remainders form the answer, with the first remainder as the *least significant bit (LSB)* and the last as the *most significant bit (MSB)*.

$$(43)_{10} = (101011)_2$$

|   |    |       |       |
|---|----|-------|-------|
| 2 | 43 |       |       |
| 2 | 21 | rem 1 | ← LSB |
| 2 | 10 | rem 1 |       |
| 2 | 5  | rem 0 |       |
| 2 | 2  | rem 1 |       |
| 2 | 1  | rem 0 |       |
|   | 0  | rem 1 | ← MSB |

# Repeated Multiplication-by-2 Method

- To convert **decimal fractions** to binary, **repeated multiplication by 2** is used, until the fractional product is 0 (or until the desired number of decimal places). The carried digits, or *carries*, produce the answer, with the first carry as the MSB, and the last as the LSB.

|                           | Carry |      |
|---------------------------|-------|------|
| $0.3125 \times 2 = 0.625$ | 0     | ←MSB |
| $0.625 \times 2 = 1.25$   | 1     |      |
| $0.25 \times 2 = 0.50$    | 0     |      |
| $0.5 \times 2 = 1.00$     | 1     | ←LSB |

$$(0.3125)_{10} = (.0101)_2$$

# Repeated Multiplication-by-2 Method

## Integer

- Division by base of target no. system
- Remainders are accumulated
- By division we obtain LSB to MSB

## Fraction

- Multiplication by base of target no. system
- Integers are accumulated
- By multiplication we obtain MSB to LSB

# Base-R to Decimal Conversion

\*\*\*Formula=  $\sum \text{digit} * \text{source\_base}^{\text{position}}$

**$(1101.101)_2$**

$$= 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-3}$$

$$= 8 + 4 + 1 + 0.5 + 0.125$$

$$= (13.625)_{10}$$

**$(572.6)_8$**

$$= 5 \times 8^2 + 7 \times 8^1 + 2 \times 8^0 + 6 \times 8^{-1}$$

$$= 320 + 56 + 2 + 0.75$$

$$= (378.75)_{10}$$

**$(2A.8)_{16}$**

$$= 2 \times 16^1 + 10 \times 16^0 + 8 \times 16^{-1}$$

$$= 32 + 10 + 0.5$$

$$= (42.5)_{10}$$

**$(341.24)_5$**

$$= 3 \times 5^2 + 4 \times 5^1 + 1 \times 5^0 + 2 \times 5^{-1} + 4 \times 5^{-2}$$

$$= 75 + 20 + 1 + 0.4 + 0.16$$

$$= (96.56)_{10}$$

# Binary-Octal/Hexadecimal Conversion

- **Binary → Octal**: Partition in groups of 3

$$\underbrace{(10)}_2 \underbrace{111}_3 \underbrace{011}_3 \underbrace{001}_3 . \underbrace{101}_3 \underbrace{110}_3 \underbrace{)}_2 = (2731.56)_8$$

- **Octal → Binary**: reverse

$$(2731.56)_8 = (10 \ 111 \ 011 \ 001 . 101 \ 110)_2$$

- **Binary → Hexadecimal**: Partition in groups of 4

$$\underbrace{(101)}_4 \underbrace{1101}_4 \underbrace{1001}_4 . \underbrace{1011}_4 \underbrace{1000}_4 \underbrace{)}_2 = (5D9.B8)_{16}$$

- **Hexadecimal → Binary**: reverse

$$(5D9.B8)_{16} = (101 \ 1101 \ 1001 . 1011 \ 1000)_2$$

# Exercise

(1) Try converting this to

$(10110001101011.111100000110)_2$

a) octal

b) hexadecimal

(2) Try converting these to binary

a)  $(673.124)_8$

b)  $(306.D)_{16}$

# Exercise Answers

1. a)  $(26153.7406)_8$

b)  $(2C6B.F06)_{16}$

2. a)  $(110\ 111\ 011\ .\ 001\ 010\ 100)_2$

b)  $(0011\ 0000\ 0110\ .\ 1101)_2$

# Exercise

$$(1054)_6 = (?)_{15}$$

Solution:

$$\text{Step 1: } (1054)_6 = (?)_{10}$$

$$\Rightarrow 1 \times 6^3 + 0 \times 6^2 + 5 \times 6^1 + 4 \times 6^0$$

$$\Rightarrow 250$$

$$\text{So, } (1054)_6 = (250)_{10}$$

# Exercise

Step 2:  $(250)_{10} = (?)_{15}$

So,  $(250)_{10} = (11A)_{15}$

|    |          |       |
|----|----------|-------|
| 15 | 250      |       |
| 15 | 16 rem A | ← LSB |
| 15 | 1 rem 1  |       |
|    | 0 rem 1  | ← MSB |

$(1054)_6 = (11A)_{15}$

# Exercise

$$(10A1B)_{13} = (?)_{18}$$

Solution:

$$\text{Step 1: } (10A1B)_{13} = (?)_{10}$$

$$\Rightarrow 1 \times 13^4 + 0 \times 13^3 + A \times 13^2 + A \times 13^1 + B \times 13^0$$

$$\Rightarrow 30275$$

$$\text{So, } (10A1B)_{13} = (30275)_{10}$$

# Exercise

Step 2:  $(30275)_{10} = (?)_{18}$

|    |            |       |
|----|------------|-------|
| 18 | 30275      |       |
| 18 | 1681 rem H | ← LSB |
| 18 | 93 rem 7   |       |
| 18 | 5 rem 3    |       |
|    | 0 rem 5    | ← MSB |

So,  $(30275)_{10} = (537H)_{18}$

$(10A1B)_6 = (537H)_{18}$

|    |   |
|----|---|
| 10 | A |
| 11 | B |
| 12 | C |
| 13 | D |
| 14 | E |
| 15 | F |
| 16 | G |
| 17 | H |

# Binary Addition: Addition Rules w/Carries

## For 2 bit

→  $0 + 0 = 00$

◆ ( 0 with a 0 carry )

→  $0 + 1 = 01$

◆ ( 1 with a 0 carry )

→  $1 + 0 = 01$

◆ ( 1 with a 0 carry )

→  $1 + 1 = 10$

◆ ( 0 with a 1 carry )

## For 3 bit

→  $0+0+0 = 00$

◆ (0 with 0 carry)

→  $0+0+1 = 01$

◆ (1 with 0 carry)

→  $0+1+1 = 10$

◆ (0 with 1 carry)

→  $1+1+1 = 11$

◆ (1 with 1 carry)

# Adding Binary Numbers

$$\begin{array}{r}
 28 \\
 + 4 \\
 \hline
 3 \\
 71
 \end{array}
 \quad
 \begin{array}{l}
 \longrightarrow \\
 \longrightarrow
 \end{array}
 \begin{array}{r}
 \textcolor{red}{0111000} \\
 00011100 \\
 + 00101011 \\
 \hline
 01000111
 \end{array}$$



# Exercise

(3) Add  $(101101)_2$  with  $(100111)_2$

# Exercise Solution

(3)  $(1010100)_2$

Working:

Augend: 101101

Addend: +100111

---

Sum: 1010100

# Addition of base-r

Example:

$$(34)_5 + (41)_5 + (24)_5$$

2 1

$$\begin{array}{r} (3\ 4)_5 \\ (4\ 1)_5 \\ + (2\ 4)_5 \\ \hline \end{array}$$

---


$$(2\ 0\ 4)_5 \quad (\text{Ans})$$

$$4+1+4=9$$

$$9\%5=4 \text{ (sum)}$$

$$9/5=1 \text{ (carry)}$$

$$1+3+4+2=10$$

$$10\%5=0 \text{ (sum)}$$

$$10/5=2 \text{ (carry)}$$

# Exercise

(4) (a)  $(FF)_{16} + (F1)_{16}$

(b)  $(66)_7 + (55)_7$

## Solution

(a)  $(1F0)_{16}$

(b)  $(154)_7$

# Binary Multiplication

- The multiplication of two binary numbers can be carried out in the same manner as the decimal multiplication.
- Unlike decimal multiplication, only two values are generated as the outcome of multiplying the multiplication bit by 0 or 1 in the binary multiplication. These values are either 0 or 1.
- The binary multiplication can also be considered as repeated binary addition. Therefore, the binary multiplication is performed in conjunction with the binary addition operation.

# Exercise

(5) Multiply 1011 with 101

|                 |               |
|-----------------|---------------|
| Multiplicand    | 1011          |
| Multiplier      | <u>x 101</u>  |
| Partial Product | 1011          |
|                 | 0000X         |
|                 | <u>1011XX</u> |
| Product         | 110111        |

# Multiplication with base-r

$$\begin{array}{r}
 2A3C \\
 \times B7 \\
 \hline
 127A4 \\
 1D094X \\
 \hline
 1E30E4
 \end{array}$$

Working

$$7 * C = 7 * 12 = 84 = (\text{write } 4, \text{ carry } 5)$$

$$7 * 3 + 5 = 26 = 1A (\text{write } A, \text{ carry } 1)$$

$$7 * A + 1 = 71 = 0x47 (\text{write } 7, \text{ carry } 4)$$

$$7 * 2 + 4 = 18 = 0x12$$

This completes the  $7 * 2A3C = 127A4$  partial product.

$$B * C = 11 * 12 = 132 = (\text{write } 4, \text{ carry } 8)$$

$$B * 3 + 8 = 11 * 3 + 8 = 41 = (\text{write } 9, \text{ carry } 2)$$

$$B * A + 2 = 11 * 10 + 2 = 112 = (\text{write } 0, \text{ carry } 7)$$

$$B * 2 + 7 = 11 * 2 + 7 = 29 = 1D$$

This completes the  $B[0] * 2A3C = 1D094[0]$  partial product, where I'm noting the [0] digits to remind us this is in the 16s column.

Adding the partial products:  $127A4 + 1D0940$

$$4 + 0 = 4$$

$$A + 4 = E$$

$$7 + 9 = 16 = 0x10 (\text{write } 0, \text{ carry } 1)$$

$$2 + 0 + 1 = 3$$

$$1 + D = E$$

$$1 = 1$$

# Exercise

(6) (a) Multiply  $(34)_5$  with  $(42)_5$

(b) Multiply  $(25)_9$  with  $(36)_9$

## Solution

(a)  $(3133)_5$

(b)  $(1033)_9$

# Binary Subtraction: Rules w/Carries

For 2 bit

- $0 - 0 = 00$  ( 0 with a 0 carry )
- $1 - 1 = 00$  ( 0 with a 0 carry )
- $1 - 0 = 01$  ( 1 with a 0 carry )
- $0 - 1 = ?$  ( ??? )

# Binary Subtraction: Rules w/Carries

$$\begin{array}{rcl}
 2 & \rightarrow & 10 \\
 - 1 & \rightarrow & - 01 \\
 \hline
 1 & \rightarrow & 01
 \end{array}$$

$$\begin{array}{rcl}
 4 & \rightarrow & 100 \\
 - 1 & \rightarrow & - 001 \\
 \hline
 3 & \rightarrow & 011
 \end{array}$$

# Exercise

(7) (b) Subtract  $(100111)_2$  from  $(101101)_2$

Solution

$(000110)_2$

Working

Minuend:      101101

Subtrahend: -100111

---

Difference:    000110

# Binary Subtraction: Rules w/Carries

$$\begin{array}{r}
 \text{3 16} \\
 (4 \text{ A } 6)_{16} \\
 -(1 \text{ B } 3)_{16} \\
 \hline
 (2 \text{ F } 3)_{16}
 \end{array}$$

$$\begin{array}{r}
 \text{4 6} \\
 (5 \text{ 4})_6 \\
 -(3 \text{ 5})_6 \\
 \hline
 (1 \text{ 5})_6
 \end{array}$$

# Exercise

$$(8) \text{ (a) } (71)_8 - (56)_8$$

$$(b) (21)_3 - (12)_3$$

## Solution

$$\text{a) } (13)_8$$

$$\text{b) } (2)_3$$

# Division of base-r

Dividend:  $x_1x_2x_3...x_m$ ; Divisor:  $y_1y_2...y_n = y$ ;

Perform  $101_2 / 10_2$ ; Dividend = 101 and Divisor = 10

Step1: Start with the first digit of the dividend. ( $x_1$ )

Step2: Compare it with the divisor;

If (it is smaller, than the divisor): ( $x_1 < y$ )

2.a append 0 to the quotient.

else: ( $x_1 \geq y$ )

proceed with division.

2.a Find the largest multiple of the divisor that fits.

2.b Append the multiplier to the quotient.

Step3: Subtract and bring down the next digit.

Step4: From now on compare the newly formed number  
in step3 with the divisor while repeating step2.

Step5. Repeat this step 2, 3 & 4 until all digits of the dividend  
( $x_1x_2x_3...x_m$ ) are processed.

**Iteration 1**

$$\begin{array}{r} 10 \overline{) 101} \\ \underline{0} \phantom{0} \\ 10 \phantom{0} \end{array}$$

**Iteration 2**

$$\begin{array}{r} 10 \overline{) 101} \phantom{01} \\ \underline{0} \phantom{0} \\ 10 \phantom{0} \\ \underline{-10} \\ 0 \end{array}$$

**Iteration 3**

$$\begin{array}{r} 10 \overline{) 101} \phantom{010} \\ \underline{0} \phantom{0} \\ 10 \phantom{0} \\ \underline{-10} \\ 01 \phantom{0} \\ \underline{-0} \\ 1 \end{array}$$

# Exercise

(9) Perform  
 $10111012 / 1102$ ; Find  
 the quotient and  
 remainder

Solution

Dividend

Divisor  $110$

$$\begin{array}{r}
 1011101 \\
 \underline{0} \\
 10 \\
 \underline{-0} \\
 101 \\
 \underline{-0} \\
 1011 \\
 \underline{-110} \\
 1011 \\
 \underline{-110} \\
 1010 \\
 \underline{-110} \\
 1001 \\
 \underline{-110} \\
 011
 \end{array}$$

Quotient  $0001111 = 1111$

Remainder  $011$

Alt. Solution

$$\begin{array}{r}
 1111 \\
 110 \overline{) 1011101} \\
 \underline{-110} \\
 1011 \\
 \underline{-110} \\
 1010 \\
 \underline{-110} \\
 1001 \\
 \underline{-110} \\
 11
 \end{array}$$

# Exercise

(10) Perform  $234102_5$   
 $/ 321_5$ ; Find the  
 quotient and  
 remainder.

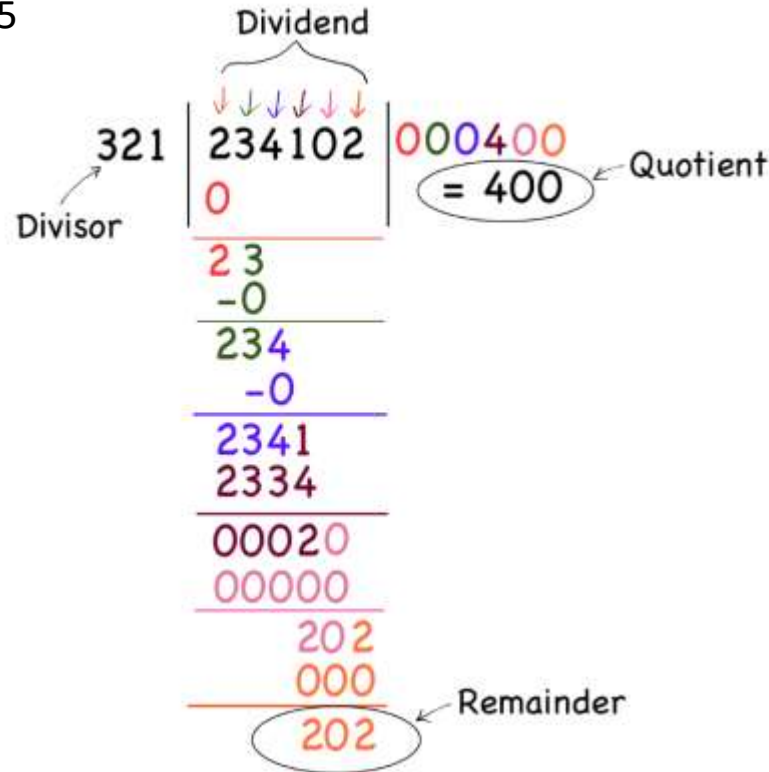
Solution

Dividend

Divisor

Quotient

Remainder



234102

0

23

-0

234

-0

2341

2334

00020

00000

202

000

202

$$\begin{aligned}
 321 \times 0 &= 0 \\
 321 \times 1 &= 321 \\
 321 \times 2 &= 1142 \\
 321 \times 3 &= 2013 \\
 321 \times 4 &= 2334
 \end{aligned}$$

# Exercise

(11) (a)  $(71)_8 / (56)_8$                       (b)  $(21)_3 / (12)_3$

Find the quotient and remainder.

Solution

Do it yourself & Compare the answer with your peers.